



PUMP FOUNDATION

PURPOSE

This document establishes guidelines for the vibration check of small pump foundations. Design consideration for the pressure forces, potential anchor forces if any, surge/hammer system forces, operating pressures and pressure fluctuations shall be included as appropriate in the final design under static checks (e.g soil bearing, concrete design, and reinforcement design etc.)

SCOPE

This document provides the following:

- A brief description of different types of pumps.
- A recommended design procedure for a soil supported foundation.
- Reinforcing requirements.
- Recommendations for grouting of pump bases.
- A chart showing coefficient of subgrade reaction K values for various types of soils (Attachment 01).
- A typical foundation sketch for a vertical (can) pump (Attachment 02).
- Design curves and example (Attachment 03 and 04)

APPLICATION

IMPERIAL UNITS: This document should be used to design a soil supported foundation for small (Mass pump skid smaller than 5000 lbs and a frequency bigger than 1200 RPM) centrifugal, reciprocating and vertical pumps. Foundations for large centrifugal pumps (greater than 500 horsepower), reciprocating pumps (greater than 200 horsepower), more than one pump supported on one mat, inline pumps and any important pump shall be designed in accordance with the guidelines listed in the Reference Section towards the end of this document.

METRIC UNITS: This document should be used to design a soil supported foundation for small (Mass pump skid smaller than 2500 KG and a frequency bigger than 1200 RPM) centrifugal, reciprocating and vertical pumps. Foundations for large centrifugal pumps (> 400 kW), reciprocating pumps (> 150 kW), more than one pump supported on one mat, inline pumps and any important pump shall be designed in accordance with the guidelines listed in the Reference Section towards the end of this document.

TYPES OF PUMPS

Centrifugal Pump

A centrifugal pump is a rotating equipment item with an impeller rotating around the shaft. The 3 main components are the pump casing, a motor, and a baseplate. These pumps are normally well balanced and do not have large unbalanced forces that create foundation vibration.



PUMP FOUNDATION

Reciprocating Pump

A reciprocating pump is a rotating equipment item with varying numbers of pistons creating a thrust. The 3 main components are the pump casing, a motor, and a baseplate. These pumps have a varying magnitude of unbalanced forces; and in general, the design is treated like a minicompressor. The foundation should be sized to avoid resonance.

Vertical (Can) Pump

A vertical (can) pump is an equipment item with a propeller revolving around the vertical axis and a housing extending below the base. These pumps are supported vertically in a pit or in a pipe casing as shown in Attachment 02. Vertical pumps are centrifugal, and their foundations do not require engineering analysis.

In-Line Pump

An in-line pump is a very small vertical centrifugal pump generally supported by the piping system. A small concrete base, as shown in Attachment 05 In-line Pump Foundations, is often provided to eliminate weight imposed on the piping system. Also reference, Pipe Supports attached to Pier Foundations Drawing 000.215.5330, which is often provided to eliminate weight imposed on the piping system.

Spring Mounted Pump

A spring mounted pump is either a centrifugal or a reciprocating pump with a base supported on springs which in turn rest on a concrete foundation. These pumps are used for very inflexible piping systems and are very rarely used. Foundation design for this type of pump should be in accordance with the recommendations of the spring isolator manufacturer.

DESIGN (SOIL SUPPORTED FOUNDATION)

General

This document outlines a recommended design procedure for a soil supported pump foundation.

To size the pump foundation, comply with the following parameters:

- Provide required edge distance for anchor bolts.
- Extend foundation a minimum of 2 inches [50 mm] around the pump base frame or the oil pan, if one exists.
- Extend the required depth below the frost line where applicable or 2 feet [600 mm] below the high point of finish surface elevation.



PUMP FOUNDATION

- Provide a mass ratio of 3 to 1 for centrifugal pumps and 5 to 1 for reciprocating pumps.
- For large pumps, obtain the anchor forces, surge forces or any other upset condition forces.
- Check soil bearing for pump weight + any other static forces using a limit of ½ the normal static allowable bearing pressure.

The rational method for sizing the pump foundation is to provide the minimum parameters listed above, check soil bearing and then to calculate the frequency ratio to confirm that it is outside the resonance range.

Typically the frequency ratio shall be outside the range of 0.7 to 1.4. Because most pumps operate between the speed of 1,200 to 3,600 RPM (revolutions per minute), the frequency ratio of 1.4 or more is recommended to avoid resonance and to keep the foundation undertuned.

The frequency ratio R is defined as the ratio of pump operating speed F , RPM, to the natural frequency F_n of pump-foundation-soil system in CPM (cycles per minute).

The natural frequency of the pump-foundation-soil system varies with the modulus of soil K that supports the foundation and the total weight at the soil surface.

Attachment 01 lists the textbook values of the modulus of soil K values. However, the Project Soil Engineer should verify the value applicable for the type of soil at the project site before using the value in the design.

The total weight of the pump-foundation-soil system includes the weight of the pump (including motor and baseplate), the weight of the foundation, and the weight of the soil. The weight of the pump, motor, and baseplate is given by the Supplier on the pump outline drawing. In most cases, the information given on the Supplier catalog sheets varies substantially from the information given on the certified outline. Use the latest certified outline.

<i>IMPERIAL</i>	<i>METRIC</i>
The natural frequency of the pump-foundation-soil system in the vertical direction is calculated: $f_n = \frac{1}{2\pi} \left[\frac{k}{M} \right]^{1/2} \text{ CPS}$ f_n = natural frequency [1/s]	The natural frequency of the pump-foundation-soil system in the vertical direction is calculated as a single mass-spring system: $f_n = \frac{1}{2\pi} \times \sqrt{\frac{k}{m}} \text{ CPS}$



PUMP FOUNDATION

substituting for k , stiffness, and M , mass

$$k = K \times A, \quad M = W / g$$

$$f_n = \frac{1}{2\pi} \left[\frac{K(12)^3 A g}{W_p + W_c} \right]^{1/2}$$

where

K = Soil modulus, pci

A = Base area of concrete, sq. ft.

g = 32.2, ft/sec/sec

W_p = Pump weight, pounds

W_c = Concrete weight, pounds

substituting

$$f_n = \frac{1}{2\pi} \left[\frac{K(12)^3 A(32.2)}{W_p + W_c} \right]^{1/2}$$

$$f_n = 37.54 \left[\frac{KA}{(W_p + W_c)} \right]^{1/2} =$$

$$37.54 \sqrt{K/P_g}$$

where:

Gross soil bearing,

$$P_g = (W_p + W_c) / A \quad \text{psf}$$

For the easy comparison the formula is rewritten for one minute and ground pressure using:

$$F_n = f_n \times 60 \text{ CPM}$$

$$F_n = 37.54 \times 60 \sqrt{K/P_g} = 2252.5 \sqrt{K/P_g}$$

f_n = natural frequency [1/s]

Where:

k = spring constant [kN/m]

m = Mass [kG],

The spring constant is derived from the soil modulus and the footing area:

$$k = K \times A$$

Where:

K = Soil modulus [kN/m³]

$$f_n = \frac{1}{2\pi} \times \sqrt{\frac{K \times A}{m}}$$

The total mass is the mass of the pump and foundation:

$$m.g = (W_p + W_c)$$

Substitution yields:

$$f_n = \frac{1}{2\pi} \times \sqrt{\frac{K \times A \times g}{W_p + W_c}}$$

For the easy comparison the formula is rewritten for one minute and ground pressure using:

$$F_n = f_n \times 60 \text{ CPM}$$

$$P_g = \frac{W_p + W_c}{A}$$

$$F_n = \frac{60}{2\pi} \times \sqrt{\frac{K \times g}{P_g}}$$



PUMP FOUNDATION

To avoid a resonance, pump operating speed, F should be at least equal to $1.4 F_n$ $F_{\min} = 1.4(F_n)$ $= 1.4(2252.5\sqrt{K/P_g})$ $= 3154\sqrt{K/P_g}$	$F_{\text{pump}} \geq 1.4 \times F_n$ $F_{\text{pump}} \geq \frac{1.4 \times 60}{2\pi} \times \sqrt{9.81} \times \sqrt{\frac{K}{P_g}}$ $F_{\text{pump}} \geq 42 \times \sqrt{\frac{K}{P_g}}$ Cycles per minute where: P_g = Soil bearing [kN/m²] K = Soil modulus [kN/m³]
---	---

Thus, for pumps with an operating speed of 1,200 RPM or greater, the simplified calculation steps for foundation design are as follows:

- Select soils modulus K , pci [kN/m³]
- Estimate minimum concrete dimensions
- Calculate total weight
- Calculate gross soil bearing, psf [kN/m²]
- Calculate $F_{\min} = 3154\sqrt{K/P_g}$ [$= 42 \times \sqrt{\frac{K}{P_g}}$]
- Compare pump speed $F(\text{RPM})$ with $F_{\min}$

If pump speed $F(\text{RPM})$ is equal to or more than $F_{\min}$, the foundation is acceptable.

If pump speed $F(\text{RPM})$ is less than $F_{\min}$, increase foundation dimensions and recalculate $F_{\min}$ until pump speed $F(\text{RPM})$ is equal to or more than $F_{\min}$.

Horizontal wind and seismic loads are not usually considered in the design of the pump foundation.

The above procedure only accounts for the vertical translation mode of vibration. Critical equipment foundations should be checked for the other vibration modes.



PUMP FOUNDATION

REINFORCING

Generally, pump foundations do not need reinforcing for stress requirements. Placing nominal reinforcing such as #5 at 12 inches [12 mm c/c 150 mm] each way is sufficient to prevent cracking of concrete surface at the top. The normal practice in most offices is to place bars on the top and all faces. In this case, provide #5 at 12 inches [12 mm c/c 150 mm] U-bars each way at top and enclose them with #4 U-ties [10 mm U-ties] on sides. Pump foundations placed on piles should be evaluated individually for reinforcing requirements.

In any event, the minimum steel requirements of ACI should be followed.

GROUTING

In general, 3 types of grout are used for grouting the pump base. These are sand-cement, epoxy, and nonshrink cementitious, or a combination of them. Sometimes the type of grout to be used is specified on Supplier outlines. In this case, in order not to jeopardize the pump warranty, use of the specified grout is recommended. Client preference plays a very important role in selecting the grout for pump bases and should be evaluated at the beginning of a project.

Pump Base

The pump base may consist of a conventional steel plate or an inverted trough fabricated using small channels around the perimeter with a steel plate on top. The cover plate supports the pump and may have holes for grouting the space between the cover plate and the concrete foundation. In either case, proper grouting of the pump bases is essential for the long term, successful operation of the pump.

Sand Cement

Sand cement grout is a mixture of 3 parts sand and 1 part cement in a trowelable consistency. This type of grout is not commonly used for grouting the pump bases.

Epoxy Grout

Epoxy grout is a mixture of resins with an exothermic characteristic; it is nonshrink in nature. Although expensive, because of its flowable characteristic, this grout fills all voids under the base plate. The result is a reliable pump performance. If the grout thickness exceeds 4 inches [100 mm], the amount of heat generated requires taking special precautions as specified by the manufacturer to eliminate the possibility of cracking the base plate.



PUMP FOUNDATION

Nonshrink Cementitious Grout

Nonshrink cementitious grout is commonly used for pump bases. As the name indicates, the grout is a cement base mixture that is compatible with concrete. It has a nonshrink characteristic when cured and leaves no voids under the base due to shrinkage. This type of grout is flowable, non-exothermic, and less expensive than epoxy grout; it is commonly used for deep pump bases.

Combinations of Grouts

Sometimes, to economize on large projects and to minimize heat generation for deep pump bases, pump bases are grouted with a combination of layers of an epoxy grout and nonshrink cementitious grout or sand cement grout. After the pump is aligned and ready for grouting, pour a layer of epoxy grout to fill the gap between the top of foundation concrete and the bottom of the base channel frame.

Once the epoxy has hardened, the space between the top of the epoxy and the underside of the cover plate is either completely filled with nonshrink cementitious grout or filled to 1 inch [25 mm] below the underside of cover plate with sand cement grout. After the sand cement grout has completely cured, pour a top layer of either an epoxy or nonshrink cementitious grout to fill the void between the second layer and the underside of the base cover plate, through the holes in the base cover plate. Another combination is to fill the base with layers of nonshrink cementitious, sand-cement, and nonshrink cementitious grouts. These procedures are labor intensive and should be evaluated with the Client and the construction representative on the project.

The grouting procedure of the pump bases is generally dictated by the Client and is defined in the project specification for grouting. Note!!! If the Client criteria differs from the PIP Guideline for Grouts 000.215.STS3600/3601, modify the project specific version of the specification.

REFERENCES

**Structural Engineering
Guideline 000.215.1233:**

Vibrating Machinery Foundations on Soil
(Using Frequency Independent Criterion)

**Structural Engineering
Guideline 000.215.1234:**

Vibrating Machinery Foundations on Soil

**Structural Engineering
Guideline 000.215.1235:**

Vibrating Machinery Foundations on Piles



PUMP FOUNDATION

Structural Engineering Guideline

000.215.5330: Pipe Supports attached to Pier Foundations drawing

Structural Engineering

Drawing 000.215.4110: In-line Pump Foundations

PIP Guideline

000.215.STS 3600/3601: Grouts

ATTACHMENTS

Attachment 01: Coefficient of Subgrade Reaction, K

Attachment 02: Typical Detail for a Vertical (Can) Pump Foundation [Imperial Units]

Attachment 03: Pump Foundation Design Curves [Imperial and Metric Units]

Attachment 04: Sample Design 1: Pump Foundations [Imperial Units]

Attachment 05: Sample Design 1: Pump Foundations [Metric Units]

Attachment 06: In-line Pump Foundations Design Criteria [Imperial and Metric Units]

Attachment 07: Pump Foundation Sizing Criteria Quick Sheet [Imperial and Metric Units]

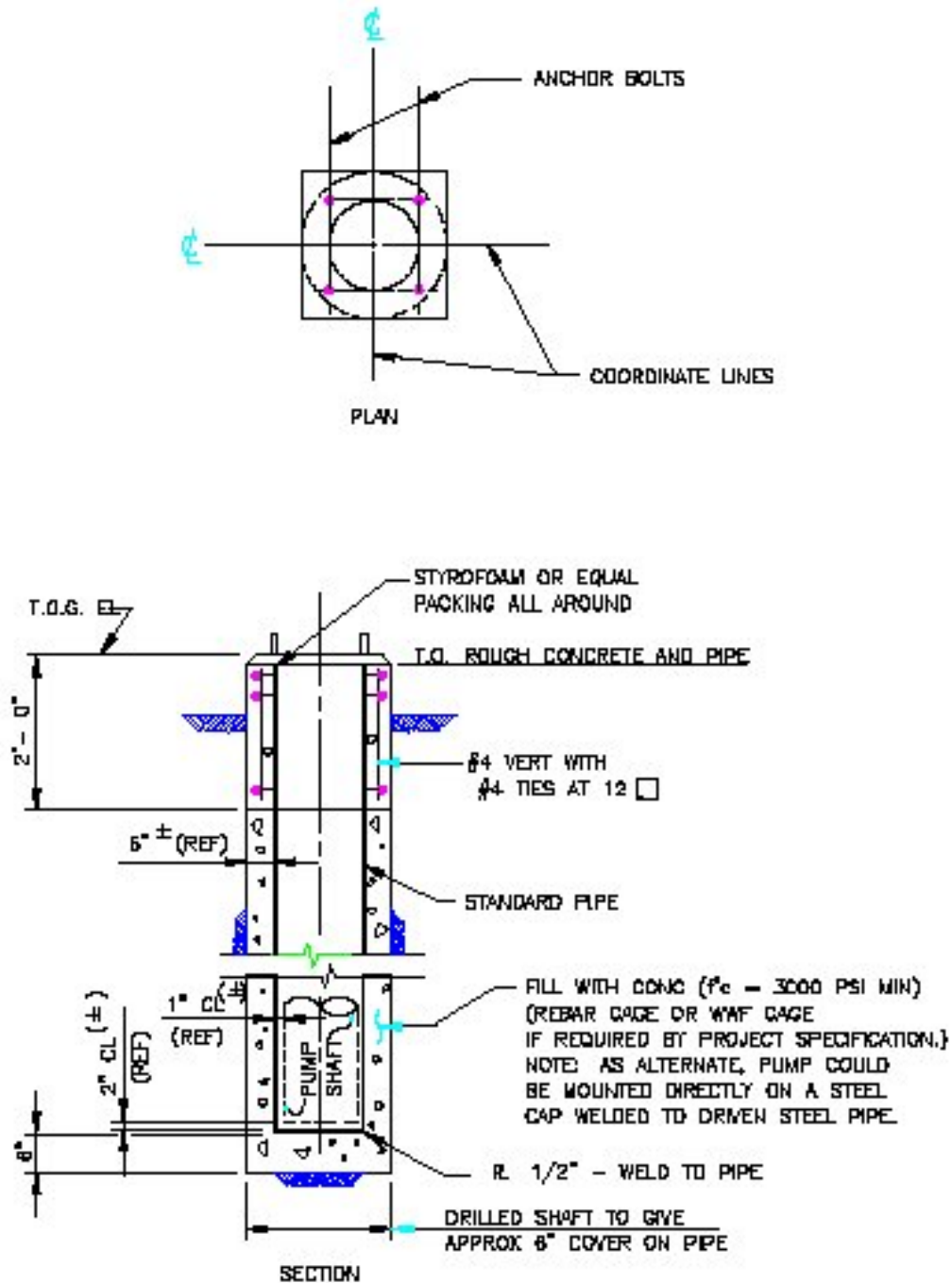


COEFFICIENT OF SUBGRADE REACTION, K

Major Divisions	Soil Groups And Typical Description	Group Symbols	K-Value PCI	K-Value kN/m ³
Gravel and Gravelly Soils	Well-graded gravel and gravel-sand mixtures. Little or no fines.	GW	500.0	135700
	Well-graded gravel-sand-clay mixtures. Excellent binder.	GC	400.0	108600
	Poorly graded gravel and gravel-sand mixtures. Little or no fines.	GP	300.0	81430
	Gravel with fines, very silty gravel, clayey gravel, poorly graded gravel-sand-clay mixtures.	GF	250.0	67860
Sands and Sandy Soils	Well-graded sands and gravelly sands. Little or no fines.	SW	250.0	67860
	Well-graded sand-clay mixtures. Excellent binder.	SC	250.0	67860
	Poorly graded sands. Little or no fines.	SP	200.0	54300
	Sand with fines, very silty sands, clayey sands, poorly graded sand-clay mixtures.	SF	180.0	48900
Fine-grained Soils Having Low to Medium Compressibility	Silts (inorganic) and very fine sands, mo, rock, flour silty or clayey fine sands with slight plasticity.	ML	150.0	40700
	Clays (inorganic) of low to medium plasticity, sandy clays, silty.	CL	130.0	35300
	Organic silts and organic silt-clays of low plasticity.	f	100.0	27100
Fine-grained Soils Having High Compressibility	Micaceous or diatomaceous fine sandy and silty soils, elastic silts.	nd	60.0	16300
	Clays (inorganic) of high plasticity, fat clays.	fa	50.0	13500
	Organic clays of medium to high plasticity.	OH	40.0	10900



TYPICAL DETAIL FOR A VERTICAL (CAN) PUMP FOUNDATION



PUMP FOUNDATION DESIGN CURVES (IMPERIAL AND METRIC)

Pump Foundation Design Curves

GIVEN:

Soil Modulus k
Pump Operating Speed RPM
Pump Weight

DETERMINE:

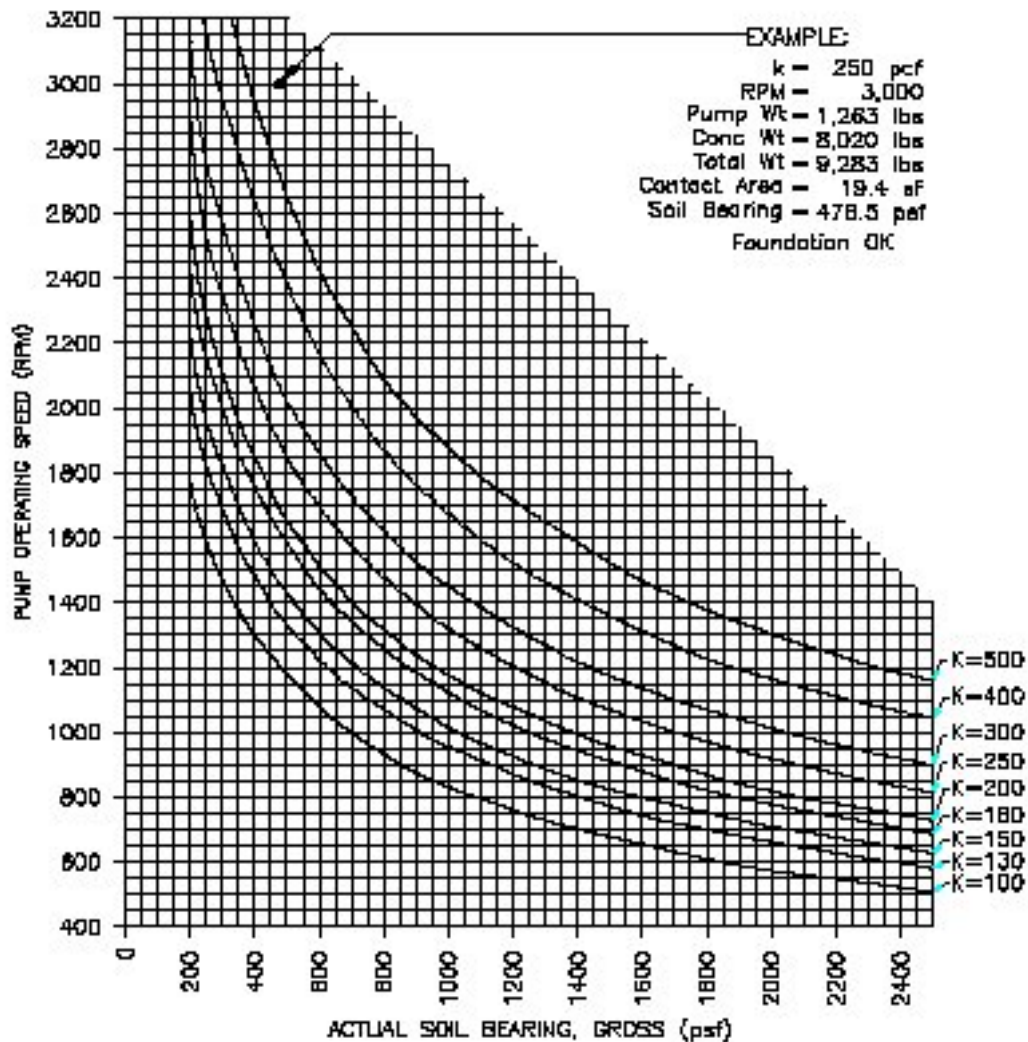
Foundation Size
Actual Gross Soil Bearing

PROCEDURE:

Plot RPM vs Soil Bearing.

1. If plotted point lies above the K -curve, foundation is OK.
2. If plotted point lies below the K -curve, deepen foundation.

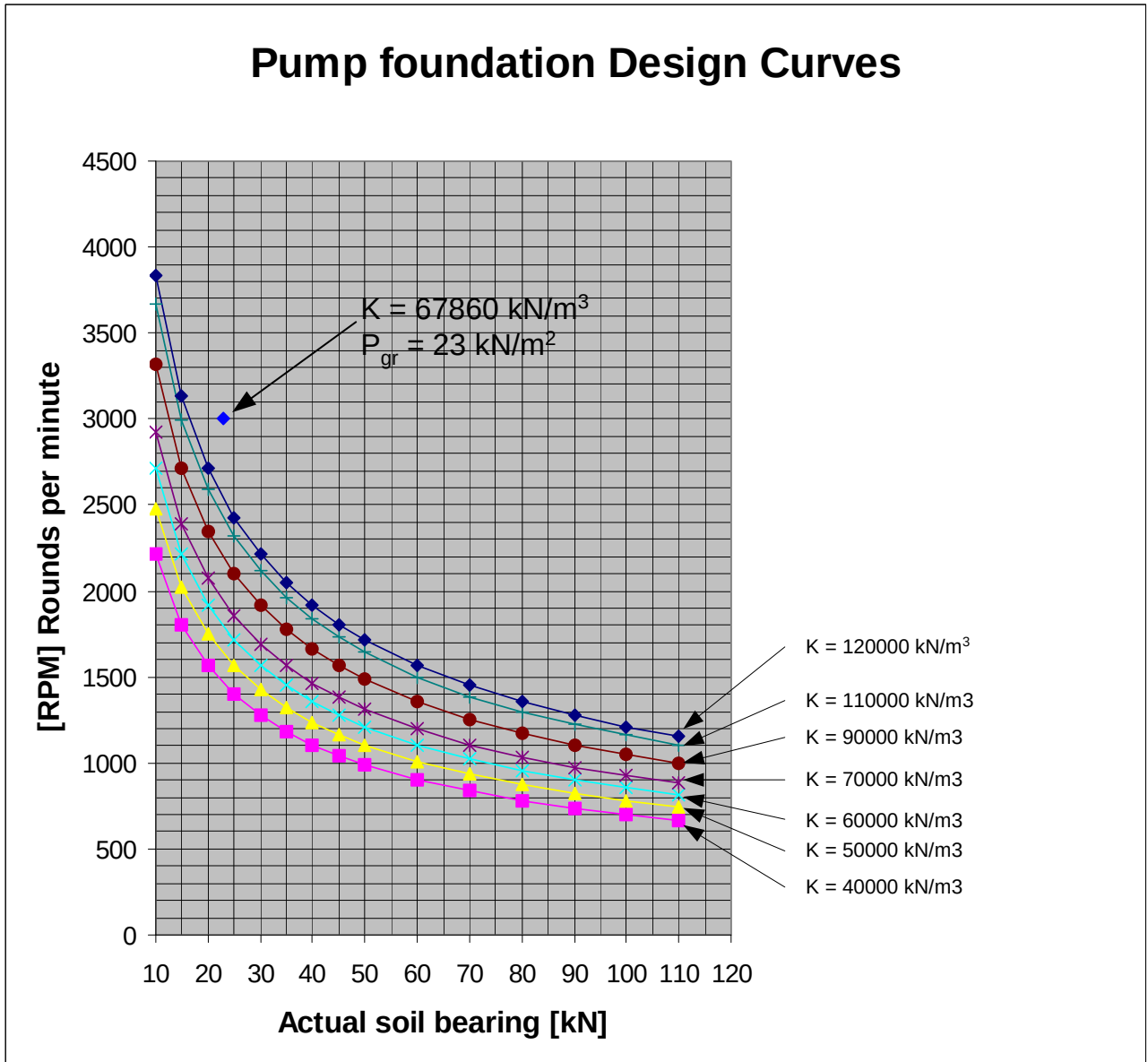
If pump operates below 1200 RPM, or is too far below the K -curve, investigate using a spread foundation similar to a reciprocating pump.





PUMP FOUNDATION DESIGN CURVES (IMPERIAL AND METRIC)

METRIC





SAMPLE DESIGN 1: PUMP FOUNDATIONS [IMPERIAL UNITS]

Design a foundation for a pump shown on attached seller's outline and for a project site with well graded sand.

1. Select coefficient of subgrade reaction from the Attachment 1
K = 250 pci for well graded sand

2. Compile minimum foundation dimensions:

Using 7/8" diameter sleeved anchor bolts for 1" diameter bolt holes.

Bolt projection = 3/8" grip (skid flange) + 1" grout + 2 1/4" nuts = 3-5/8" = say 4"

Standard bolt length = 1'-6"

c/c distance between outermost bolts is 27" x 72.5"

Minimum foundation size for bolt edge distance = (27" + 2x6x7/8) x (72.5" + 2x6x7/8") i.e. 37.5" x 83"

Pump base size is 30" x 84.5"

Minimum foundation size for oil pan protection

= (30" + 2 x 2 = 34") x (84.5" + 2 x 2 = 88.5")

← Governs →

Depth of foundation = 9" above HPFS + 2'-0" below HPFS = 2'-9" (no frost requirement)

Try foundation 38" x 90" x 2'-9" deep (3'-2" x 7'-6" x 2'-9" deep)

$W_{fdn} \text{ (concrete)} = (3.17') \times (7.5') \times (2.75') \times (150\text{pcf}) = 9,807 \text{ lbs}$

Total weight of pump, motor, and base = 2324 lbs

$W_{fdn} / W_{\text{pump tot}} = 9807 / 2324 = 4.2 > 3 \text{ ----- Ok}$

3. $F_{\text{pump speed}}$ (motor speed) = 3600 RPM (from seller outline or data sheet)

$F_n \text{ (foundation)} = 3154 \sqrt{(K/P_j)}$, where K = 250 pci and

$P_j = (2324 + 9807) / (3.17 \times 7.5) = 510 \text{ psf}$

$F_n = 3154 \sqrt{\frac{250}{510}} = 2208 \text{ CPM} < 3000 \text{ RPM ----- Ok}$

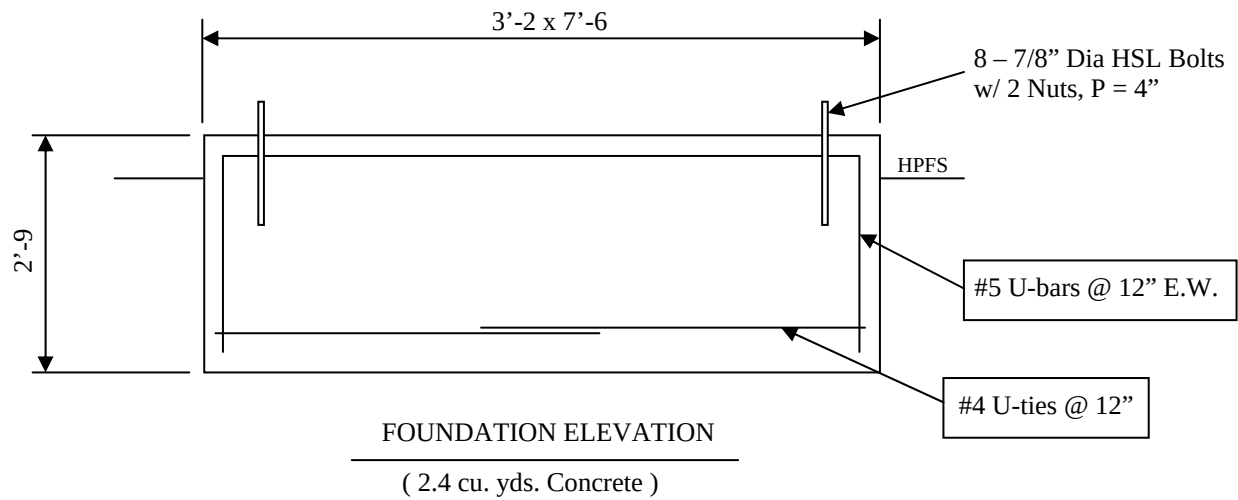
4. Reinforcing:

In lieu of any client requirements provide a minimum reinforcing # 5 at 12 c/c mat each way at top only.

5. Provide 3'-2" x 7'-6" x 2'-9" fdn. with # 5 at 12 each way at top and side
8-7/8" HSL Bolts with 2 nuts and 4" projection



SAMPLE DESIGN 1: PUMP FOUNDATIONS [IMPERIAL UNITS]





SAMPLE DESIGN 1: PUMP FOUNDATIONS [METRIC UNITS]

Design data:

Pump base size is 760 x 2150 mm
c/c distance between outermost bolts is 700 x 1800 mm
Soil constant K = 67860 kN/m³ for well graded sand

1. Compile minimum foundation dimensions:

Using 22 mm diameter sleeved anchor bolts for 25 mm diameter bolt holes.
Bolt projection = 10 mm grip (skid flange) + 25 mm grout + 60 mm nuts = say 100 mm
Standard bolt length = 450 mm
c/c distance between outermost bolts is 700 x 1800 mm
Minimum foundation size for bolt edge distance = (700 + 2x6x22) x (1800 + 2x6x22) i.e. 960 x 2060 mm.

Pump base size is 760 x 2150 mm
Minimum foundation size for oil pan protection
= (760 + 2 x 50 = 860 mm) x (2150 + 2 x 50 = 2250 mm) ← Governing

Depth of foundation = 250 mm above HPP + 600 mm below HPP = 850 mm (no frost requirement)

Try foundation 1000 mm x 2300 mm x 900 mm deep.

$W_{fdn} \text{ (concrete)} = (1,0) \times (2,3) \times (0,9) \times (2500 \text{ KG}) = 5175 \text{ KG}$

Total weight of pump, motor, and base = 1100 KG

$W_{fdn} / W_{pump \text{ tot}} = 5175 / 1100 = 4.7 > 3 \text{ ----- Ok}$

2. F_{pump} speed (motor speed) = 3000 RPM (from seller outline or data sheet)

$F_n \text{ (foundation)} = 42 \times \sqrt{(K/P_{gr})}$, where K = 67860 kN/m³ and

$P_{gr} = (5175 + 1100) (10) / (1,0 \times 2,3 \times 1000) = 27 \text{ kN/m}^2$

$F_n = 42 \times \sqrt{\frac{67860}{27}} = 2106 \text{ CPM} < 3000 \text{ RPM ----- Ok}$

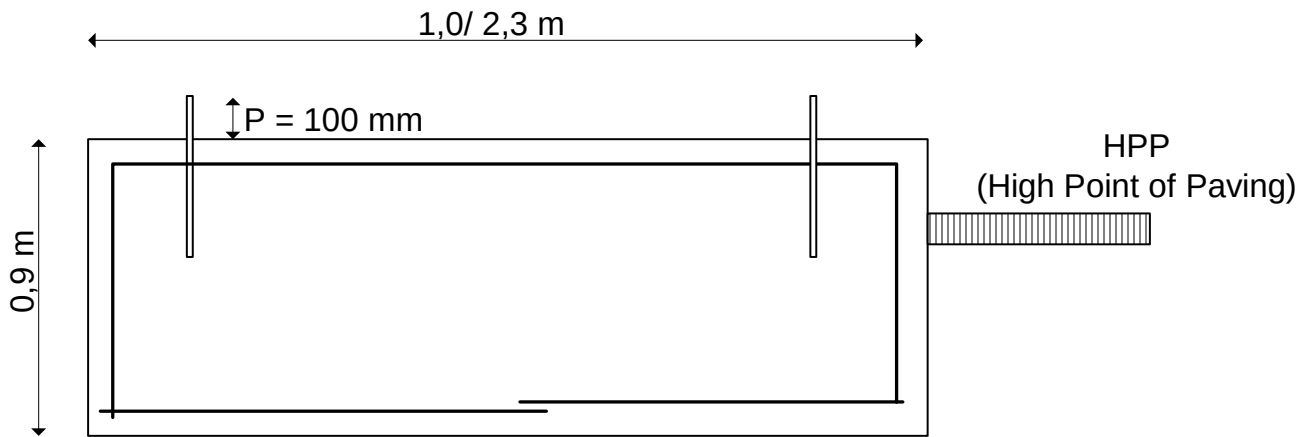
3. Reinforcing:

In lieu of any client requirements provide a minimum reinforcing 16 mm at 300 mm c/c mat each way at op only.

4. Provide 1 m x 2,3m x 0,9m fdn. with 12 mm at 150 mm each way at top and side 8- 22mm HSL Bolts with 2 nuts and 100 mm projection



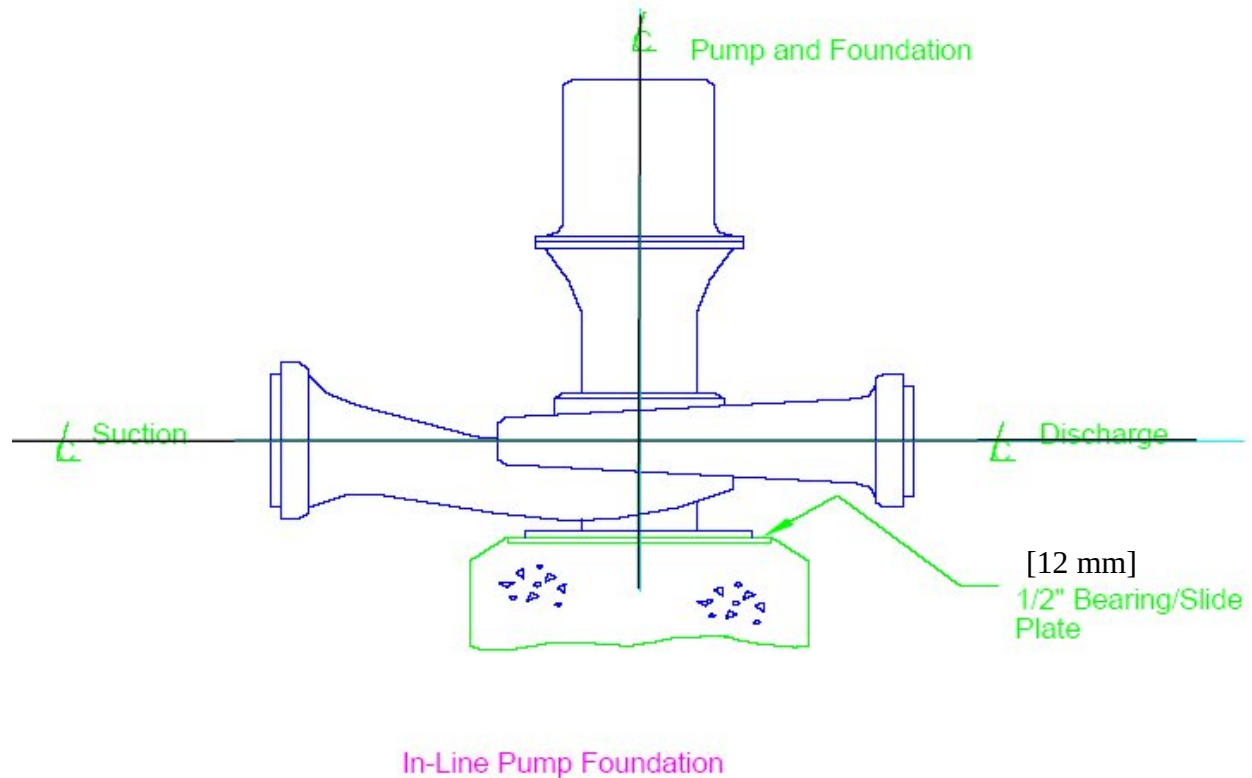
SAMPLE DESIGN 1: PUMP FOUNDATIONS [METRIC UNITS]



Foundation Elevation



IN-LINE PUMP FOUNDATION DESIGN CRITERIA



Notes:

1. Pump to be line supported except where total pump and motor weight exceeds 250 times the nominal size of the discharge nozzle [100 times the nominal size in inches of the discharge nozzle].

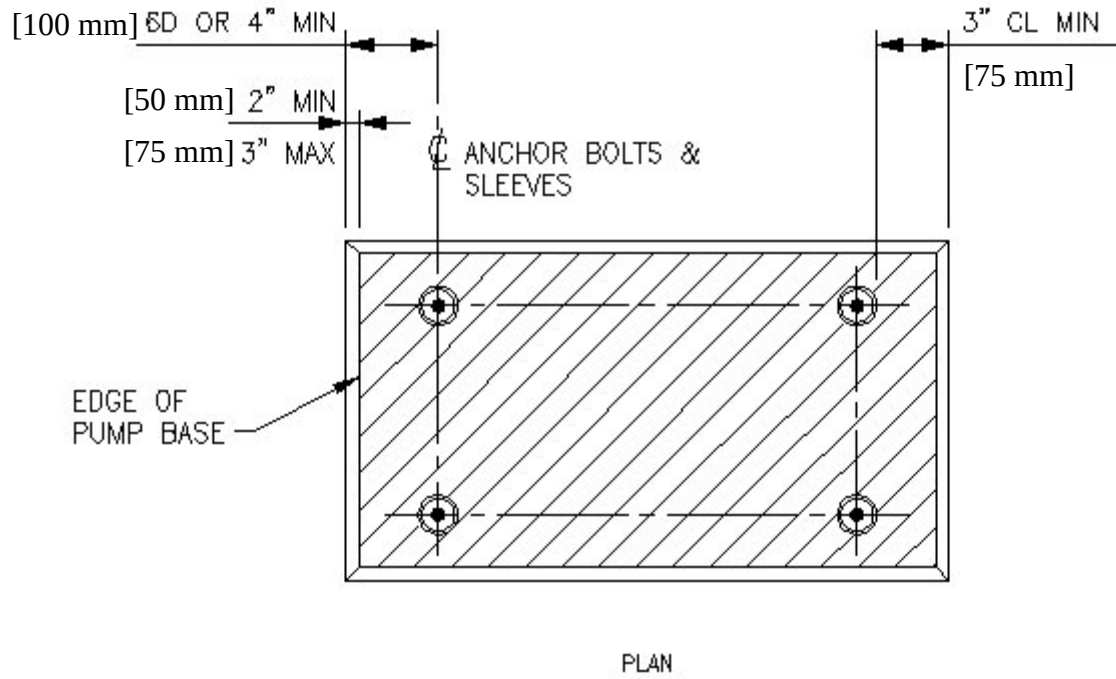
Example:	Imperial	Metric
Pump and motor weight =	1500 lbs	680 kg
4" Discharge nozzle		
	4 x 250 = 1000 lbs	4 x 100 = 400 kg
	1500 lbs > 1000: Support required	680 kg > 400 Support required

2. No fastening between pump and slide plate or between concrete and slide plate required.
3. Coordinate lines for all in-line pumps shall be centerline of pump, not centerline of discharge.
4. Edge of foundation shall extend a minimum of 2" and a maximum of 3" beyond the pump based.
5. Bearing/slide plate shall extend 1" minimum beyond the pump base.
6. Overall dimensions of foundation (width and length) should be rounded up to the nearest even inch.



PUMP FOUNDATION SIZING CRITERIA QUICK SHEET

- 1.0 Determine pump foundation size from Manufacturer's outline with the following rules.
 - 1.1 Edge of foundation shall extend a minimum of 2 inches [50 mm] and a maximum of 3 inches [75 mm] beyond the pump base, 4 inches [100 mm] beyond the centerline of anchor bolts, 3 inches [75 mm] from the face of bolt sleeves, or 6 bolt diameters, whichever is greatest. (The 2-inch [50 mm] dimension allows grout to be poured more easily; the 3-inch [75 mm] maximum dimension, if practical, prevents grout cracking which normally occurs at bolt locations).
 - 1.2 Overall dimensions of foundation (width and length) should be rounded up to the nearest even inch [100 mm].
 - 1.3 Generally, pump foundations are set 2'-0" [600 mm] below HPFS unless frost line requires a deeper foundation or a pile cap is required. Pump Foundations are sometimes extended to pick up adjacent pipe supports.
 - 1.4 Normally, top of grout is set 9" [250 mm] above HPFS.
- 2.0 When the combined weight of the pump, motor and frame exceeds 5,000 lbs [2250 kG], the foundation must be checked for vibration.





PUMP FOUNDATION SIZING CRITERIA QUICK SHEET

GIVEN:

Soil Modulus K
Pump Oper. RPM
Pump Weight
Pump Fdn. (per page 1 of this practice, Practice 000.215.1237 and Job Requirements)

DETERMINE:

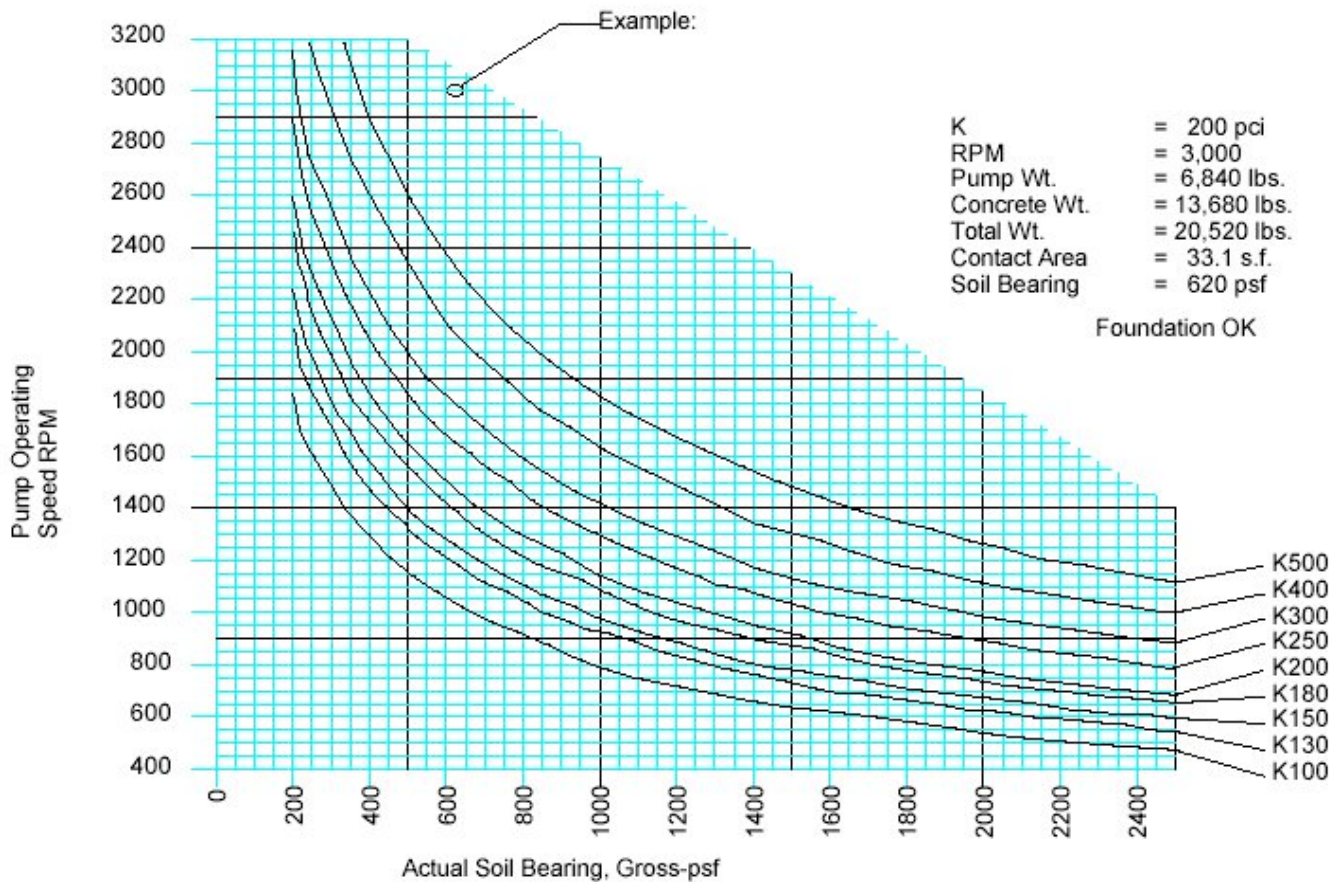
Actual Gross Soil Bearing

PROCEDURE:

Plot RPM vs. Soil Bearing.

1. If Plotted Point lies above the K-curve, foundation OK.
2. If Plotted Point lies below the K-curve, deepen foundation.

If pump operates below 1200 RPM or is too far below the K-curve, investigate using a spread foundation similar to a reciprocating pump.





PUMP FOUNDATION SIZING CRITERIA QUICK SHEET

